

WDD (EGL204) & OSWSD (EGL205)

Project Requirements for EGL204 & EGL205

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Individual Project Development (IPD) Brief

Academic Integrity

Plagiarism is the act of taking someone else's work or ideas, whether in whole or in part, and presenting them as your own without acknowledging the original source.

Nanyang Polytechnic (NYP) takes plagiarism seriously and considers it a form of academic dishonesty. Learner who is found to have violated NYP Academic Integrity Policy by committing an act of plagiarism during a module assessment, will be subjected to disciplinary action.

Learners would be liable for disciplinary action if they were found to provide or seek unauthorized assistance such as:

- Copying another learner's answer
- Providing another learner with answer
- Fabricating data, information or citations
- Asking another person to complete the work that you're supposed to do
- Taking and using the whole or any part of ideas, words or works of others and passing it off as your own work without acknowledgement of the original source (including AI resource)

Please refer to [NYP Academic Integrity Policy](#) in the NYP website for the treatment on contravention of academic integrity.

Late submission policy

- **Submit before Deadline:**
 - **By 8th Aug 2025 (Friday), 11:59 PM**
 - **No penalty**
- **Submit late but within 5 calendar days:**
 - **From 9th Aug 2025 to 13 Aug 2025, 11:59PM**
 - **Project score capped at 50%**
- **Submit late after more than 5 calendar days:**
 - **After 13th Aug 2025 11:59PM**
 - **0 mark for project score**

WDD (EGL204) & OSWSD (EGL205)

1. Objectives

In this IPD phase, you will be developing a web application for booking events (e.g., concerts, movies screening, workshops) using HTML, CSS, JavaScript, jQuery, PHP, and MySQL, showcasing concepts learned in EGL204 and EGL205.

The grading for EGL204 will be based on the UI/UX design as well as the level of user intuitiveness & interaction, while grading for EGL205 will be based on overall application design and implementation of functional requirements.

2. UI/UX Design Requirements

Wireframing and Page Flows

Typically, UI/UX designers will minimally go through the process of web application design such as wireframing and page/screen flows. Hence, you will be required to develop wireframes and screen flows of all the pages used in the application. You may draw or sketch out the wireframes and screen flows and submit them in PowerPoint slides. **Only wireframes of the Desktop format are required to be submitted.**

Responsive Web Design

The web application will need to support Responsive Web Design (which uses Media Query) to determine the screen's dimension. You will code the appropriate behaviour of the pages according to the device's screen width. Your web application will need to support the following device types and typical breakpoint pixel value ranges:

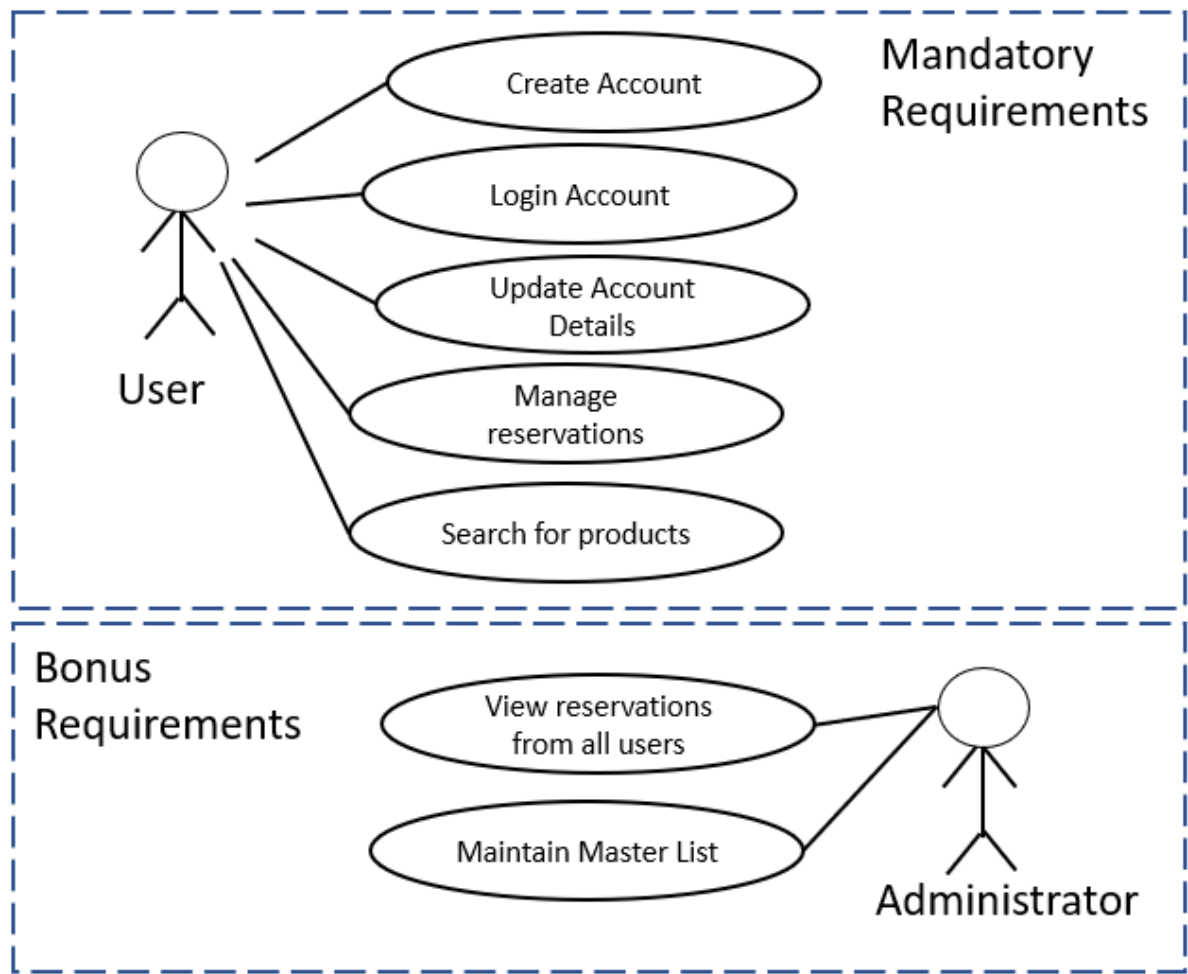
- 1) Mobile phone (typically < 480px screen width, with 1 column of content)
- 2) Tablet (typically >= 768px screen width with 2 columns of content)
- 3) Desktop screens (typically >= 1200px or more with 3 or 4 columns of content)

You are allowed to make suitable adjustments on the actual RWD breakpoints according to the contents of the web page, but these breakpoints should be consistently applied on all web pages once the optimal breakpoints have been determined.

Use of 3rd party CSS/JavaScript Framework(s) not permitted

To allow for fair assessment of students' competence in fundamental skills of web application development, the use of 3rd party CSS frameworks (e.g., Bootstrap, Foundation, etc), open-source icon libraries (e.g. font awesome, ion icons, etc), and JavaScript frameworks (e.g., React, Angular, Vue, etc) will not be permitted. Use of these frameworks will incur heavy penalties. Only the use of jQuery is allowed. Please approach your tutor if you require any clarifications.

3. System Requirements Specification



3.1 Home Page/Landing Page

Requirement Ref	Requirement Specifications
3.1.1 Home/ Landing Page Description	The home page/landing comprises of the following information at minimum: a) Logo for the App b) Navigation Bar to facilitate user to access the various functionalities c) Hero image/Web banner to publicise the upcoming events. d) Footer with copyright information

WDD (EGL204) & OSWSD (EGL205)**3.2 Create Account**

Requirement Ref	Requirement Specifications
3.2.1 Account Fields	Users will be able to create free accounts with the following fields: a) User ID b) Email Address c) Password d) Mobile number Upon successful account creation, the user will be brought to the login page.
3.2.2 Validation	Client Validation via JavaScript/jQuery during account creation: a) Email Address must be of the correct format b) Password Passwords must be at least 8 characters long and not more than 16 characters in length c) User are required to re-type password during account creation that matches the initial password. d) Mobile number must contain only 8 digits, beginning with 8 or 9. Server Validation during account creation: a) UserID has to be unique across all accounts in the system b) Email address does not exist in the database

3.3 Login

Requirement Ref	Requirement Specifications
3.3.1 Login	Users can only log in using their User ID and password. Upon successful login, the user will be directed to the event listings page, which displays all upcoming events available for booking.

WDD (EGL204) & OSWSD (EGL205)**3.4 Update Account**

Requirement Ref	Requirement Specifications
3.4.1 Validation	User can only update their account details after they have successfully logged in. User can only update the following fields: a) Email Address b) Mobile Number The new Email address cannot be one that is already present in the database belonging to an existing account.

3.5 Manage Reservations

Requirement Ref	Requirement Specifications
3.5.1 Events Page	Upon successful login, the user will land on the event listings page, which displays all available events Users can book an event by selecting a specific event and providing the required details, such as the number of attendees and the event date. Users can search for events using a case-insensitive search. (For example, searching "music festival," "Music Festival," or "MUSIC FESTIVAL" will return the same set of results from the database)
3.5.2 Reservations Page	The user will be able to view all their booked events. The user can cancel an event booking at any point of time.

WDD (EGL204) & OSWSD (EGL205)**3.6 View reservations from all users (Bonus)**

Requirement Ref	Requirement Specifications
3.6.1	Only the administrator can view all event bookings made by all users.

3.7 Maintain Master List (Bonus)

Requirement Ref	Requirement Specifications
3.7.1	The administrator will be able to insert, update, and delete items in the system's master list of events.

3.8 Security Requirements

Besides the landing page, the application shall NOT allow direct access to individual webpages without the user having first logged in. In such scenario, the application will redirect user to the login page.

4. Assessment Considerations**UI/UX & Web Design Assessment Considerations (EGL204)**

1. Design of screen and page flows that leads to intuitive use of the web application
2. Consistent use of suitable fonts, colours schemes and good contrast based on principles and knowledge learnt in UXD.
3. Proper design and implementation of Responsive Web Design (RWD) in all pages
4. Proper layout, spacing and sizing of web page elements, considering RWD behaviours.
5. Suitable level of interactivity using JS/jQuery to improve user experience and web application functions

Application Development Assessment Considerations (EGL205)

1. Database tables and fields are appropriately designed to support the application
2. Proper setup of database accounts for application access
3. Meeting all the functional requirements of the web application

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5. Submission Deadlines

Project Part 1 (EGL204 10%)

- Wireframes for all pages (only Desktop format)
- Page flows for all pages

Mode of submission: Preferably in a PowerPoint format. Other formats (e.g, Adobe XD, Figma, etc) are acceptable also. Please discuss with your teacher.

Submit by end of Week 8 for review.

Project Part 2 (EGL204 30% & EGL205 40%)

- Fully functional responsive and interactive web application
- Source codes
- Database dump

Due: Week 16, Friday, 2359 hours.

Refer to Brightspace for the assessment rubrics for each component.

6. Reference Resources

- a. Adobe Colour Wheel: <https://color.adobe.com/create/color-wheel>
- b. Adobe Contrast Checker: <https://color.adobe.com/create/color-contrast-analyzer>
- c. Modern Fonts for Websites: <https://xd.adobe.com/ideas/principles/web-design/best-modern-fonts-for-websites/>